House Price Prediction System

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1. Problem Description

The House Price Prediction System is a machine learning-based application designed to estimate property prices based on various features such as location, size, number of rooms, and other relevant factors. In the real estate market, accurately predicting house prices is a complex task due to multiple influencing variables. This project aims to automate the prediction process using machine learning techniques, reducing human effort and improving decision-making for buyers, sellers, and investors. The system uses data preprocessing, feature selection, and regression models to learn patterns from historical housing data and generate accurate price predictions. This helps in understanding market trends and making informed financial decisions.

2. Architecture

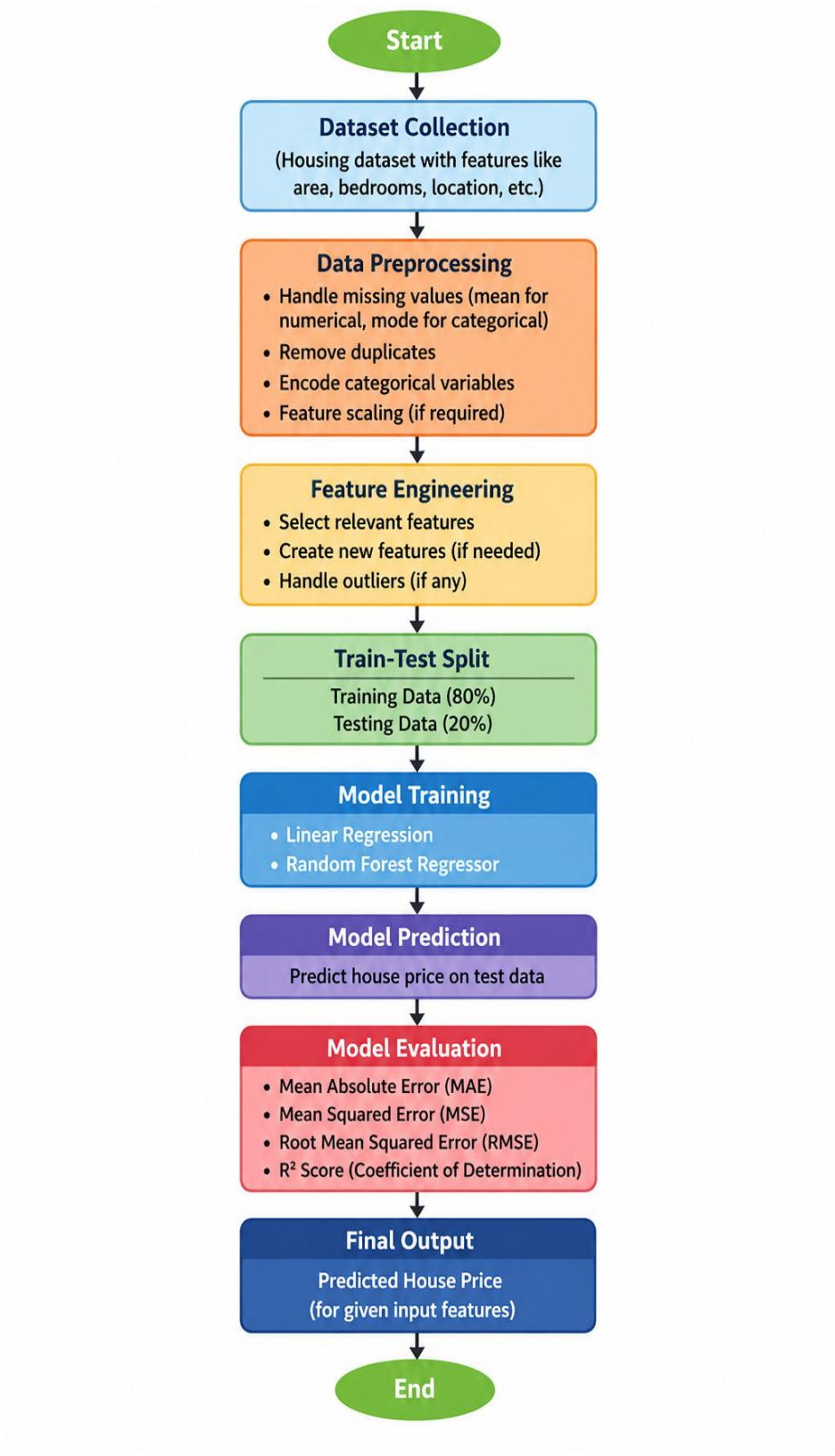

Fig.1 Sample flow diagram.

3. Models Used(Applied/ Planned) Give a brief introduction of the models used

1. Linear Regression

Linear Regression is a basic supervised learning algorithm used for predicting continuous values. It assumes a linear relationship between independent variables and the target variable (house price).

- Simple and fast
- Works well for linear data
- Easy to interpret

2. Random Forest Regressor

Random Forest is an ensemble learning algorithm that uses multiple decision trees to improve prediction accuracy.

- Handles non-linear relationships
- Reduces overfitting
- Provides better accuracy than linear models in complex datasets

4. Evaluation metrics used

To evaluate the performance of the models, the following metrics were used:

1. Mean Absolute Error (MAE)

- Measures average error
- Lower value = better model

2. Mean Squared Error (MSE)

- Penalizes larger errors more
- Helps identify model performance

3. Root Mean Squared Error (RMSE)

- Square root of MSE
- Easier to interpret in real-world units

4. R^2 Score (Coefficient of Determination)

- Measures how well the model explains variance
- Value ranges from 0 to 1
- Closer to 1 = better model

5. Preliminary Results and Visualizations

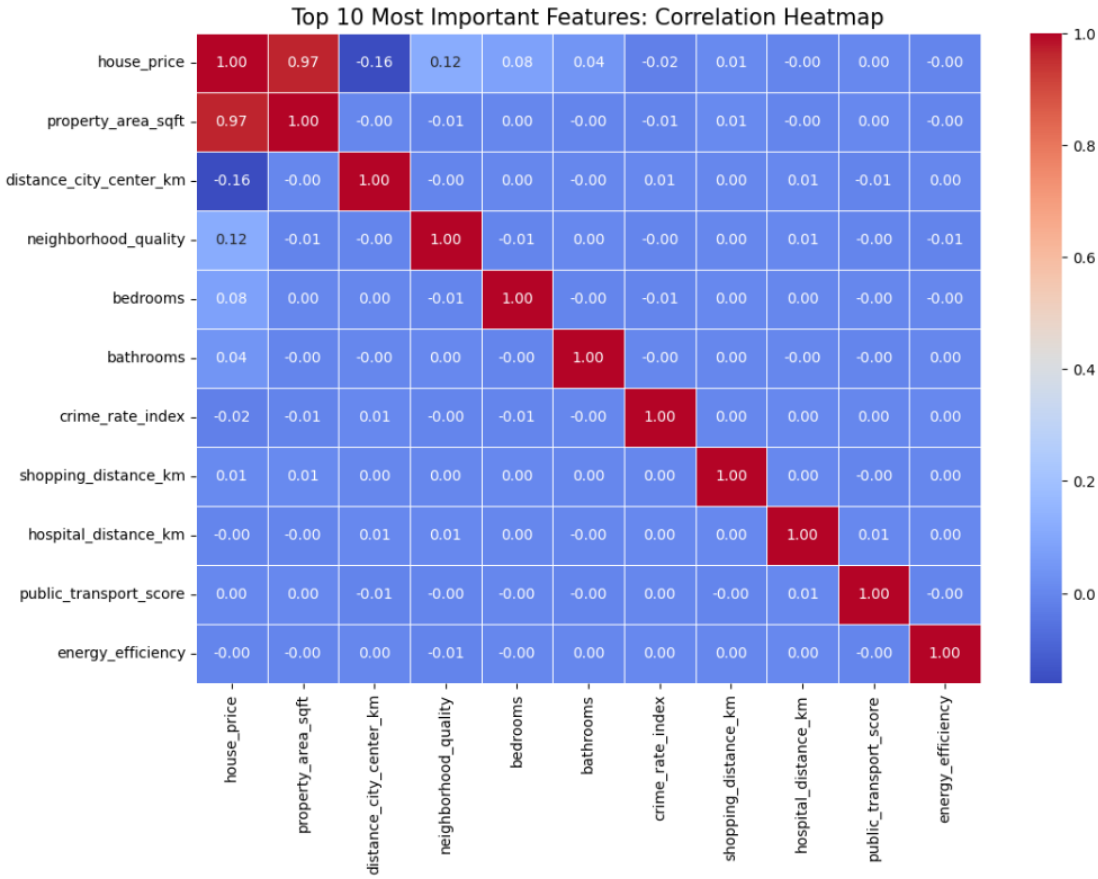

Fig.2 Heatmap showing corelation.

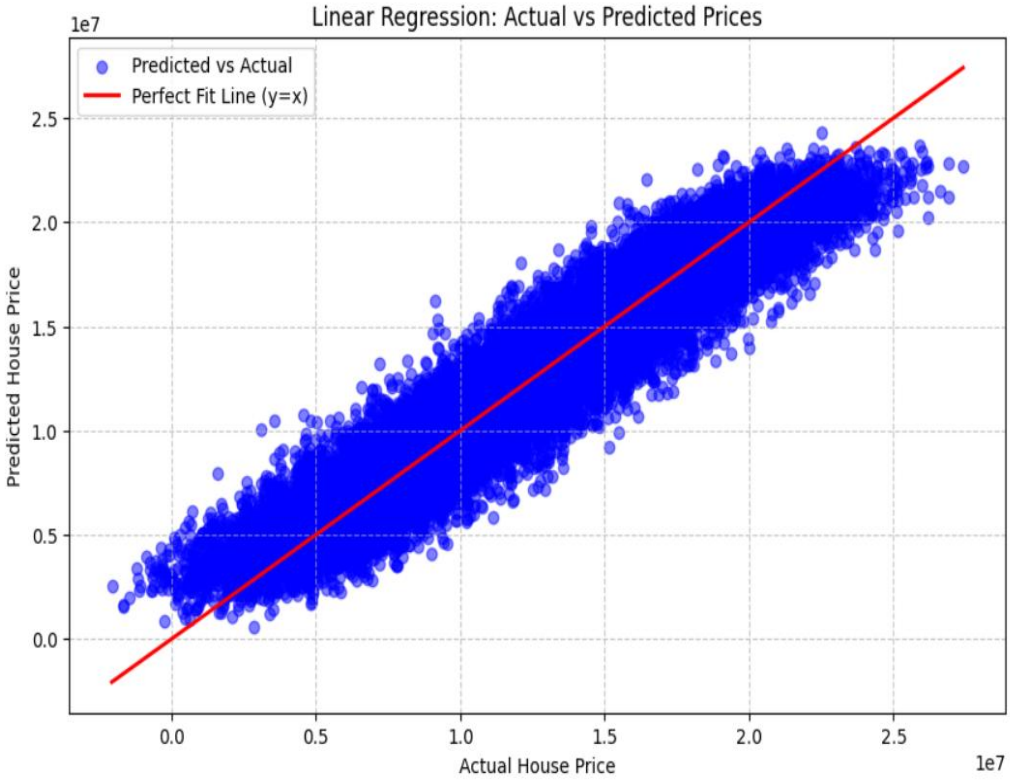

Fig.3 : Linear Regression Best fit line graph

6. References:

Books:

1. Let Us Python – Yashavant Kanetkar

Online Resources:

- GeeksforGeeks – https://www.geeksforgeeks.org/
- TutorialsPoint – https://www.tutorialspoint.com/cprogramming/index.htm
- Stack Overflow – https://stackoverflow.com
- Scikit-learn Documentation -- https://scikit-learn.org
- Chatgpt -- https://chatgpt.com/c/69e66a5a-6708-8324-97f1-428e1440bb70, https://chatgpt.com/share/69f0bc3f-70a8-83e8-8ef6-56dab4e802ad